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An Investigation into Predictive Modelling of Malaysian Stock Prices Using An Optimum Feature Set

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Abstract. The stock market is often seen as a sentiment indicator which can indirectly impact the GDP (gross domestic product) of the nation either negatively or positively. At the personal level, many may have invested their life savings in retirement funds which are managed by professional fund managers. Thus if fund managers performed badly in the stock market, the performance of stock market will indirectly impact the returns of the retirement fund under their care. However, due to the huge amount of data that they are dealing with, it can be very challenging if not humanly impossible to analyze all the data to come up with an accurate model of the performance as well as the potential of the stocks. Artificial intelligence can help improve the effectiveness of fund managers. This can then allow a fund manager to manage multiple funds at a time, which in turn lowers the fund management fees, effectively giving the client a bigger capital return. This paper describes a novel approach of using financial news sentiment and genetic algorithm to predict the performance of Malaysian stock prices. The performance of the final optimized model is also compared with other popular predictive models.

INTRODUCTION

There are two main categories of investing strategy in the financial sector which is fundamental analysis and technical analysis. The two are very distinctive as fundamental analysis focuses on the long term return while technical analysis usually focuses on short term returns. Fundamental analysis generally focuses on the fundamentals of a company. The core concept of fundamental analysis is to study the health of a company's financial statement and the future prospect of the company by studying its financial statements, business model, market competitors, the overall market, financial ratios, etc. A simple way of doing fundamental analysis is to look at a company's revenue, earnings, assets, liabilities, etc. This strategy favours big companies that have strong earning power, especially market leading companies as they have healthier cash flow and better financial ratios (e.g. return on equity, dividend yield). Generally, fundamental analysis strategy requires a longer amount of time ranging from months to years to reap profit. This is because the value of a company may not be accurately reflected in a short period of time according to this Inefficient Market Hypothesis. On the other hand, technical analysis is a strategy that identifies trading opportunities from the price trend and price fluctuations. Investors that employ technical analysis as the main strategy believe that historical price changes of a stock can be valuable indicators of the stock's future price movements [1]. Hence experts in the financial industry have identified many technical indicators that can determine market trends, strength of market, likelihood of continuation etc. over the years.

The rest of the paper is organized as follows. The following section highlights the key points of prior work. Section 3 describes the experimental framework used to evaluate the performance of the various predictive algorithms for stock price prediction. The results are shown and analyzed in Section 4. Finally, Section 5 presents the conclusions of this investigation and recommends some areas for future work.

PRIOR WORK

Rui Li et al. [2] used stock forum discussion feeds from the Shanghai Composite Index to generate the structural emotion time series of the stock market. The method used to analyze the sentiments of investors relies on comparing the number of negative words and positive words in the forum discussion. Their results had found that there is a strong interaction between the emotion index and the volatility of the stock market.

A popular algorithm for stock price prediction is ARIMA (AutoRegressive Integrated Moving Average) and Ayodele et al [3] used ARIMA to predict stock prices. Although they have not quantified their accuracy, but from the plotted graph of the actual and predicted values it can be concluded that the accuracy obtained was not satisfactory. Nelson, Pereira and Oliveira [4] had implemented LSTM (Long Short Term Memory) with price history and various technical indicators as inputs, managing to achieve an average accuracy of 55.9%. They had suggested the accuracy can be improved by having better data preprocessing and using a new set of features. Furthermore they suggested that the key to improve accuracy is to improve the quality of the input data.

Sahaj and Govinda [5] took a different approach by treating stock prediction as a classification problem with class up, down and unchanged. They used the Random Forest algorithm and Support Vector Machines (SVM) for their classification. . The features selected were news data from Reddit, Dow Jones historical price and Guardian News feed. The news was processed using a Bag of Words model and ngram model to quantify the data. The n-gram (or skip-gram) model used stores the words in the order in which they occur in the data. The Random Forest model using a 1-gram model for text analysis produced an accuracy of 84.3% and on using a 2-gram model produced an improved accuracy of 86.2%.

D. Shah [6] had investigated the effects of financial news sentiments on the stock market. The text classification algorithm used was a dictionary-based classifier. The decision to buy, hold or sell was made based on the calculated sentiment score. An accuracy of 70.59% was achieved with such a dictionary-based approach. They have also suggested to use word weighting approach in sentiment analysis to determine which word is more important and then assign the sentiment score accordingly.

C.Y. Lee [7] used recurrent convolutional neural networks (RCN) to predict future stock prices. The author used a mixture of word vectors and historical price of a variety of stock and index as input, and train the RCN over a period of time to minimize the price error to predict future prices. The text classification algorithm used was word2vec which was developed by Google Inc. The results suggested that the price prediction of RCN is better than LSTM with lower root mean square error. The root means square error of RCN varies among stock but the average root means square error of RCN is less than 6.

Kamble [8] performed short term and long term price prediction using Random Forest algorithm. The prediction problem was treated as a classification problem, with an output to buy, hold or sell. For short term stock price prediction, Kamble used moving average convergence divergence (MACD), stochastic oscillator, and Bollinger bands along with several financial ratios such as debt to equity ratio. For short term stock price prediction, around 1,000 stocks were tested on the model. 563 out of the 1,000 stocks were given a buy recommendation, with 376 of the stocks hitting 10% of the target price. The accuracy of the model was reported to be 66.8%. Kaya and Karsligil [9] took a different approach by including financial news to predict future stock prices. They retrieved financial news articles published in the previous year and get stock prices for same period as their dataset. Words and phrases consisting of nouns and verbs were used as features instead of using single words. The SVM classifier was trained with labeled train articles. The classification has only two classes which is positive and negative news. The results showed that the accuracy from this model managed to achieve only 60%. Wang, Ho and Lin [10] investigated the relationship between stock prices and news sentiment and successfully predict future stock price. To benchmark the performance they used two models, one with sentiment analysis and the other without sentiment analysis. SentiMo as well as Vader Sentiment Analyzer were used to obtain sentiment analysis results. The main algorithm used was neural network with back propagation. Their results showed that the accuracy of model with sentiment analysis performed better than the model without sentiment analysis.

A summary of the techniques used for stock price forecasting is shown in Table 1. Notice that the best results as reported came from SVM with news feeds and achieving 86.2%

TABLE 1. Summary of various techniques for stock prediction

Methodology	Feature Used	Accuracy
AutoRegressive Integrated Moving Average [3]	Not specified	Low
Recurrent Convolutional Neural Network [7]	Word vectors from financial news	50.0%
Long Short Term Memory [4]	Various technical indicators	55.9%
Support Vector Machine [10]	Labelled news articles	60.0%
Random Forest [8]	Various technical indicators	66.8%
Random Forest [5]	Financial news data	85.0%
Support Vector Machine [9]	Reddit news data and Guardian news feed	86.2%

METHODOLOGY

The general flow of building a predictive model starts by collecting data for the model to train on. Next step is to perform feature extraction after which the data must be preprocessed to fit the requirements of the predictive model. The final step is to train the predictive model with the preprocessed data. Model parameter tuning is performed until the model is performing optimally.

Data Collection

The historical price data was taken from Yahoo Finance. The historical data was in a form of .csv file, with 5 data columns comprising of the opening price, highest price, lowest price, closing price and volume traded. The financial ratios (earnings, assets) of a stock were also taken from the website of Bursa Malaysia – the stock exchange of Malaysia using a custom webscraper written in Python language. Bursa Malaysia is one of the largest bourses in ASEAN (Association of Southeast Asian Nations which comprises of ten countries in Southeast Asia). The financial news data was extracted from The Star’s Business’ Twitter account in the form of tweets. (*The Star* is Malaysia's top English-language daily). The Python library twitterscraper was used to automate this task as the number of tweets are too large for manual data collection. A total of 82,884 tweets from the year 2010 until 2019 were collected. Some of the common factors that affect stock prices include,

.Price moving averages

i. Volume

ii. News

Figure 1 illustrates the performance of the Bursa Malaysia composite index accompanied with the big news that affected it in recent years (1997 Asian financial crisis, bankruptcy of Lehman Brothers, subprime mortgage crisis, etc.)



FIGURE 1. Performance of the Bursa Malaysia index for the top 30 companies (1996 - 2018)

Feature Extraction

The categorical data were converted to numerical data by assigning a discrete value for each and every category. Besides that, all data were cleaned by removing alphabets in numbers as well as outlier values.

Eight features chosen from those that had been extensively used in many of the prior work were adopted here, viz.

i	<i>Historical Closing Price</i>	v	<i>Positive Return Days over the past 100 days</i>
	<i>Volatility Index</i>	v	<i>Negative Return Days over the past 100 days</i>
	<i>Relative Strength Index</i>	v	<i>Changes in Earnings per Share</i>
	<i>Volume Traded</i>	v	<i>Financial news sentiment</i>

Data Preprocessing

In order to take financial news as an input to a predictive model, the textual data must be quantified for the ease of computation. To automate this, the Python library NLTK (Natural Language ToolKit) was used to perform sentiment analysis of financial news data. Each piece of financial news was analyzed for its sentiment. The sentiment score ranges from -1 to 1, with -1 being negative, 0 being neutral and 1 being positive. After computing the sentiment score for every piece of financial news tweet, the mean of the sentiment score was calculated to represent the sentiment score on that particular date. The time series dataset had to be processed to fit into the predictive model. Since the goal of the predictive model is to predict the next 10 days of closing price, the model expects 10 days of historical data as input and 10 days of future closing price as label/output.

The stock prices of nine companies were chosen. The reason of choosing companies across different sectors is to eliminate selection bias as some sector might outperform other sectors at the same time frame. The sector chosen were financial service sector (*Ambank Berhad*), telecommunications sector (*Digi Berhad*), consumer products sector (*Heineken Malaysia Berhad*), healthcare sector (*IHH Healthcare Berhad*), plantation sector (*Kuala Lumpur Kepong Berhad*), logistics sector (*MISC Berhad*), electronics manufacturing sector (*Malaysian Pacific Industries Berhad*), chemical industry sector (*Petronas Chemical Berhad*) and public utilities sector (*Tenaga Nasional Berhad*).

Predictive Models Investigated

Figure 2 shows the 5 types of predictive models investigated, namely *XGBoost*, *AdaBoost*, *Multi Layer Perceptron Neural Network*, *Long Short Term Memory* and *Convolutional Neural Networks*. The reason of choosing these different models is to compare the performance and suitability of different types of models on stock prices.

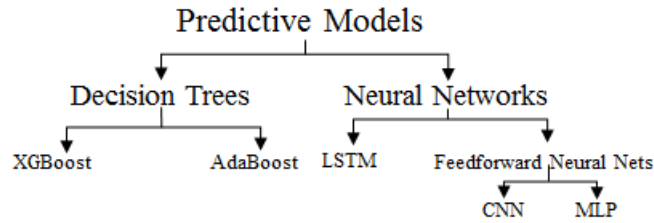


FIGURE 2. Predictive models investigated

RESULTS AND DISCUSSIONS

The performance metrics used to measure the performance of the predictive models used are Mean Absolute Percentage Error (MAPE), accuracy and *Alpha*. Alpha is a commonly used performance metric in the fund management industry. It measures the active return of an investment compared to a benchmark market index.

The dataset was analyzed using XGBoost which is a decision tree-based model to determine the feature importance of the dataset, resulting in a descending order of importance of “*Previous Closing Price*”, “*Change in Earnings*”, “*Relative Strength Index*”, “*Volatility Index*”, “*Volume*”, “*Positive Return Days*”, “*News Sentiment*” and finally “*Negative Return Days*”.

Results

Table 2 shows the performance of the various predictive models investigated here. For MAPE, a small value is desirable and this was achieved with LSTM which returned a MAPE of 4.06%.

TABLE 2. Performance for different predictive models

Performance metric	M APE (%)	Accu racy (%)	Al pha (%)	Ret urn
XGBoost	5.9 1	33.31	7. 00	1.1 8
AdaBoost	6.0 6	33.27	-0. 30	1.0 5
MLP	4.3 5	32.78	3. 00	1.1 4
CNN	4.1 0	32.83	3. 00	1.1 2
LSTM	4.0 6	31.06	-9. 00	1.0 8

On the other hand, *Alpha* which measures the active return of an investment compared to a benchmark market index a higher value is desirable. The same goes for accuracy as well as the return. XGBoost was able to give better results compared to the rest of the models investigated here. This decision tree based predictive model outperformed other neural network based predictive models when measured against these three metrics. This contradicts with the current hype of artificial neural network where it is believed to be top performing in most machine learning task. There are a few possible explanations to this finding. First, the size of dataset is relatively small with only one year of data (8,760 rows of data), neural network works better with larger amount of data whereas decision tree-based algorithms need substantially less data to work well. Secondly, neural network has many hyperparameters which needs to be tuned in order to achieve optimal performance, e.g. number of features, etc. Hence to improve the performance of the neural network-based predictive models, the MLP was optimized with genetic algorithm (GA).

GA, inspired by Charles Darwin's theory of natural evolution, reflects the process of natural selection where the fittest individuals are selected for reproduction in order to produce offspring of the next generation. The process of natural selection starts with the selection of fittest individuals from a population. They produce offspring which inherit the characteristics of the parents and will be added to the next generation. If parents have better fitness, their offspring will be better than parents and have a better chance at surviving. This process keeps on iterating and at the end, a generation with the fittest individuals will be found. Here only the best candidates were allowed to continue into the next generation. The GA was able to identify better performance for the MLP by eliminating two of the 8 features, namely "*news sentiment*" and "*negative return days*".

Table 3 shows the best performance of the predictive model on the 9 different stocks from different sectors. The optimised GA-MLP was able to deliver better results across all the four different measures.

TABLE 3. BEST Performance of Optimized Predictive Models

Performance metric	M APE	Accu racy	Al pha	Ret urn
Decision Trees	5.9 1	33.31	7. 00	1.1 8
Predictive models (MLP with GA)	3.0 4	33.70	9. 30	1.2 0

CONCLUSIONS AND FURTHER DEVELOPMENTS

This investigation compares a genetic algorithm optimized multi layered perceptron to multiple popular time series predictive models. From the findings, genetic algorithm optimized multi layered perceptron performed better than conventional, popular model such as XGBoost. This approach demonstrates the ability of genetic algorithm to select the best feature. This study only uses 8 features, further explorations can be made with a higher dimension dataset which might lead to higher accuracy.

For stock price prediction, the performance of the predictive model performed better than buying and selling every 10 days. Back-testing results also showed that an individual can actually profit from buying and selling according to the predictions made by the model. It performs well on most of the stocks chosen here. The predictive model was able to predict the stock prices with a small error of only 4.05% on average but fails to capture the direction of the price movement. It is worth to note that the predictive model was able to capture the direction of price movement during uptrend and downtrend movements. Nevertheless, it performs mediocrity when the price movement goes sideways (also known as horizontal movement when the forces of supply and demand are nearly equal).

In all of the prior work reviewed, there was no mention of the time frame used to make predictions. A shorter time frame might yield higher accuracies but the predictions might not be as useful because of the extremely small predicted time frame. Nevertheless, it would be interesting to study the impact of the time frame on the accuracy and if an optimal time frame can be identified. Other than this, more features can be studied. The financial news collected can be refined by only taking company specific news. This can remove noise in data as not every financial news was relatable to a specific company. The sentiment analysis of financial news can be improved as well by using those extracted from better Natural Language Processing methods. With their ability to discover patterns in nonlinear and chaotic systems, neural networks offer the ability to predict market directions more accurately than some of the current techniques investigated. However, there are many variants on offer and hence another area of investigation is to look at the performance from other neural network architectures.

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